

Irfan Mubarak

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SUMMARY

AI & Data Science student with a focus on Autonomous Systems. Proven expertise in building real-time computer vision pipelines and multi-agent AI frameworks. Deeply interested in contributing to research involving Secure IoV communication and Intelligent Cyber-Physical Systems.

TECHNICAL SKILLS

- **AI & Security:** Computer Vision (YOLOv8), RAG Pipelines, Secure AI Integration, LLM Orchestration
- **Systems & Frameworks:** CrewAI, LangChain, FastAPI, Flask, OpenCV, PyTorch
- **Blockchain & Networks:** Knowledge of Distributed Ledgers, Consensus Mechanisms (PBFT), REST APIs
- **Tools & Automation:** n8n, Workflow Automation, Linux, Git, Google Cloud APIs, Pinecone

KEY PROJECTS

Real-Time Multi-Modal Conversational AI System

2025

FastAPI, LangChain, Whisper STT, Pinecone, RAG

- Developed a stateful, event-driven pipeline for multi-modal interaction with sub-second latency.
- Optimized a **Retrieval-Augmented Generation (RAG)** pipeline for contextual synthesis, relevant to remote monitoring and decision-support systems.
- Implemented session management for high-concurrency environments.

TAPACD - Theater Anti-Piracy AI Camera Detection

2024

YOLOv8, OpenCV, Flask, Roboflow, Python

- Built a real-time surveillance system utilizing custom-trained **YOLOv8** for edge-case object detection.
- Achieved robust performance in low-light environments, demonstrating skills in **Intelligent Cyber-Physical Systems (CPS)** and real-time sensor monitoring.

Multi-Agent AI Decision System

2025

Python, CrewAI, LLMs, JSON Orchestration

- Designed a collaborative system using specialized agents to solve complex research and analysis tasks.
- Demonstrated expertise in **distributed task execution** and structured reasoning, applicable to collaborative AI and multi-node network security.

AI-Driven Recruitment Automation & Decision System

2026

Python, Flask, Gemini, n8n, API Orchestration

- Engineered an end-to-end automation pipeline featuring multi-source data ingestion and AI evaluation.
- Leveraged **event-driven architecture** to handle unstructured data, mimicking complex IoV data streams.

EXPERIENCE

AI Intern — Ilmora AI Solutions

Jul 2025 – Oct 2025

- Focused on industry-level model evaluation and the automation of complex workflows using n8n.
- Gained hands-on experience in integrating AI features into production-grade systems.

Student Placement Co-ordinator — MES College of Engineering

2024 – 2026

- Coordinated high-level corporate interactions and managed logistical execution for campus-wide drives.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

2022 – 2026

MES College of Engineering, Kuttippuram (APJ Abdul Kalam Technological University)